

VACUUM REFLOW SOLDERING: A Shift For Reliable Electronics

As electronics becomes increasingly complex, traditional reflow soldering is being replaced by a process that delivers cleaner solder joints, greater reliability, and improved performance in advanced circuit designs. But what exactly is vacuum reflow soldering?



PCB after full assembly moves to reflow oven machine for heating up soldering paste

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Although not yet widely adopted, vacuum reflow soldering is emerging as an important process in high-reliability electronics manufacturing. As power densities increase and component packages become smaller, conventional reflow soldering often struggles with one persistent issue: solder voids.

These microscopic gas pockets trapped within solder joints reduce thermal conductivity, weaken mechanical strength, and can ultimately compromise the long-term reliability of electronic assemblies.

How do solder voids form

Voids typically form when gases released during flux activation, trapped air, oxide

reduction, or large thermal pad geometries become trapped within molten solder during reflow. Once the solder solidifies, these gases remain trapped, creating voids that are difficult to eliminate using conventional processes.

Vacuum reflow addresses this challenge by applying a controlled vacuum while the solder is in its molten state. The reduced chamber pressure causes trapped gas bubbles to expand and escape before the solder solidifies, allowing the molten solder to flow into the resulting spaces. This results in denser solder joints with significantly lower void content, improved thermal conductivity, improved electrical performance, and greater mechanical reliability.

Inside the vacuum reflow process

Unlike conventional reflow, the vacuum is not applied throughout the entire heating cycle. The PCB first passes through preheating and heating zones until the solder melts. A short vacuum stage is then introduced before cooling begins. Applying the vacuum too early has little effect, while applying it too late allows voids to become permanently trapped. Synchronising the vacuum cycle with the solder's liquid state is therefore critical to achieving optimum results.

In conventional surface-mount technology (SMT), excessive voiding can compromise joint integrity, reduce heat dissipation, and accelerate failures under thermal cycling, vibration, and mechanical stress. Vacuum reflow helps produce solder joints that can better withstand these demanding conditions. This makes the technology particularly suitable